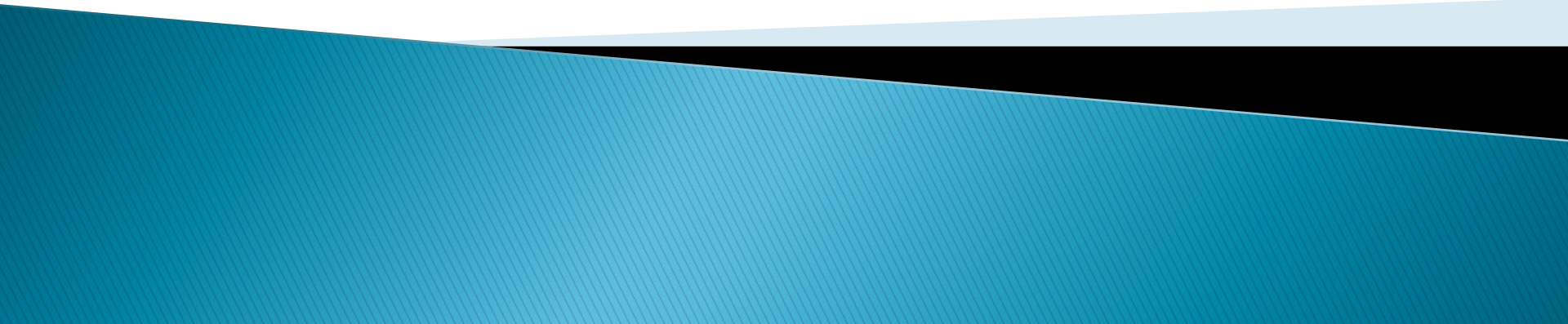


Carbon Emission Trends and Industry Impact

By Jason Dayuha

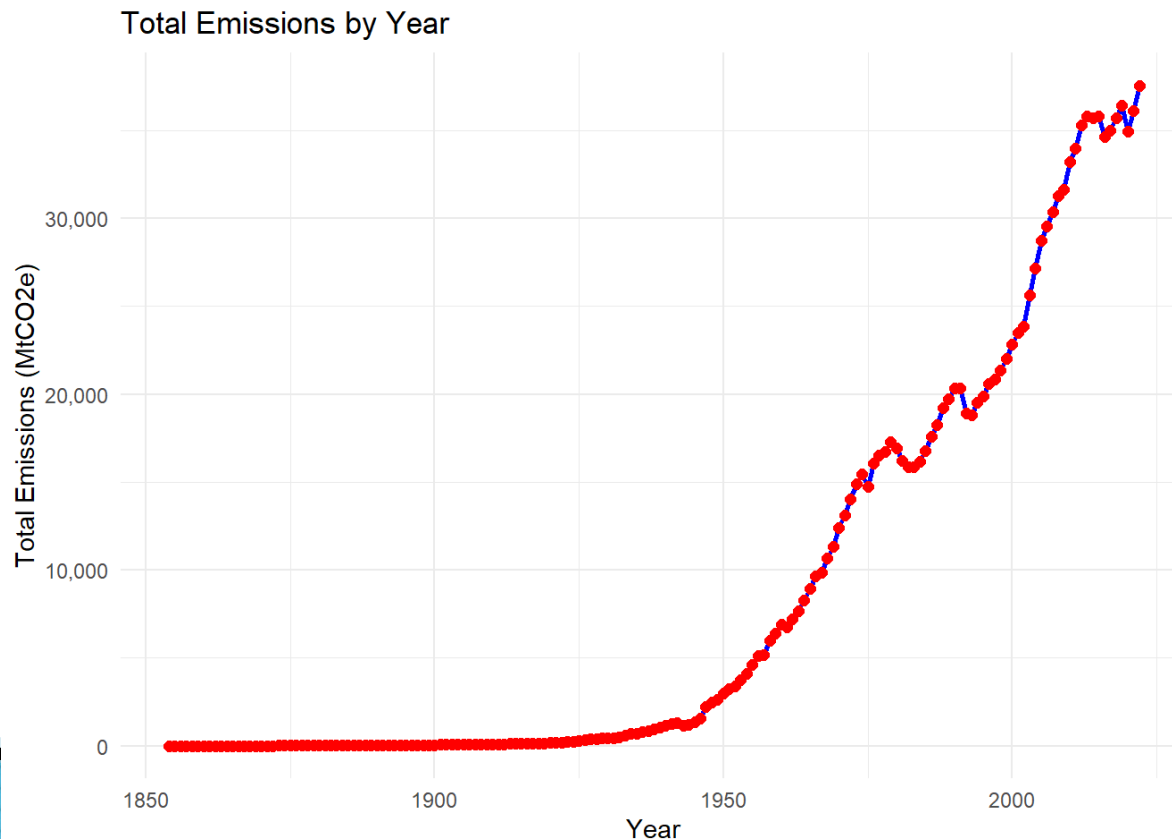


Introduction

- ▶ This project is part of the **Tidy Tuesday weekly data analysis program**, a community initiative for improving data skill.
- ▶ This report focuses on the week of 05-21-2024, specifically examining carbon major emissions production worldwide. The project explores the amount of emissions produced by the largest energy companies globally through real datasets.

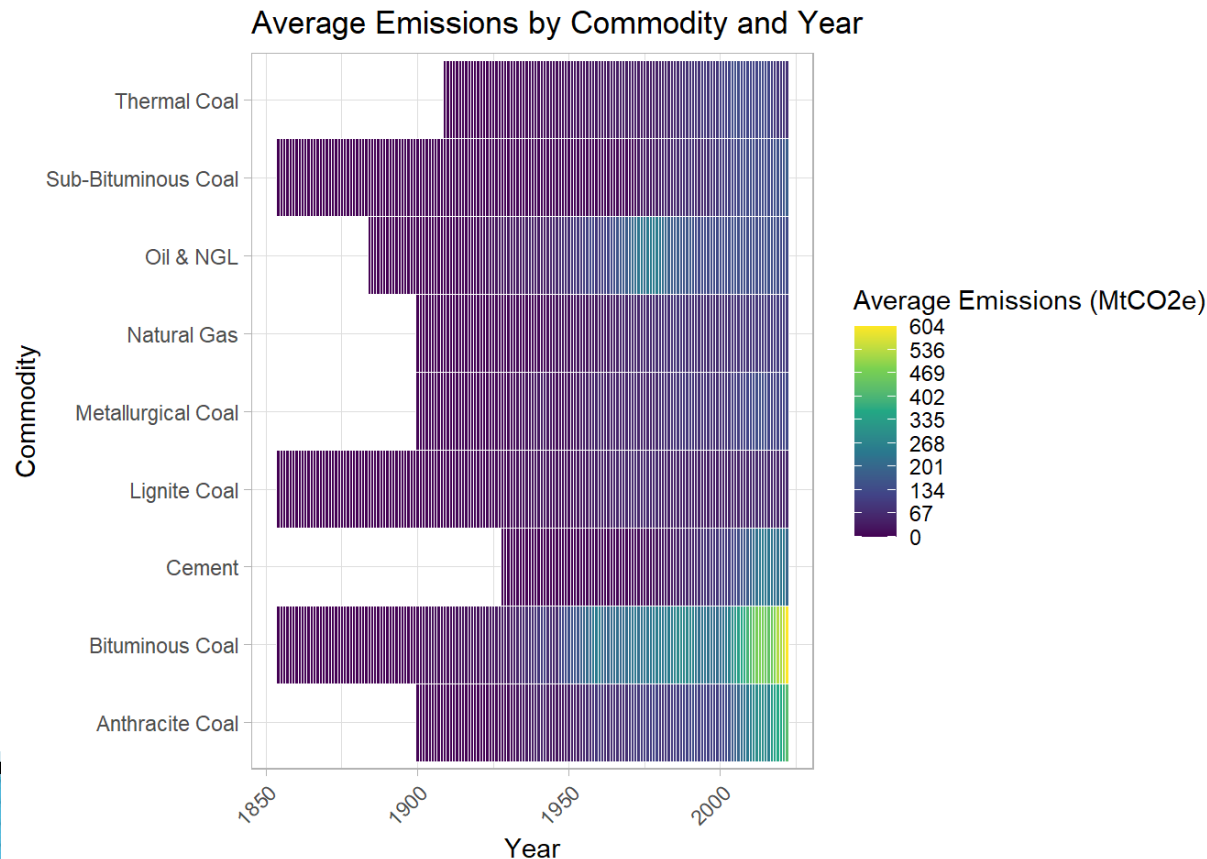
Global Emissions Over Time

- ▶ **Total emissions** rose steadily from 1854 to the 1940s, then **sharply increased** to present day.
- ▶ Slight dips occurred in the 1980s–1990s, but the overall trend remains upward.
- ▶ By 2022, global emissions reached **record highs**, showing continued dependence on fossil fuels.



Major Emitters & Commodity Trends

- ▶ Top contributors include China (coal), Former Soviet Union, and Saudi Aramco.
- ▶ China leads with 276,000 MtCO₂e, followed by the Former Soviet Union at 135,000 MtCO₂e.
- ▶ **Coal-based commodities**—bituminous, lignite, and sub-bituminous—drive the majority of emissions.
- ▶ Emissions from coal peaked between 500–600 MtCO₂e around 2021–2022, while oil and cement show smaller but steady increases.



Production, Intensity, and Entity Insights

- ▶ Higher **production values** generally align with **higher total emissions**, especially for coal-related commodities.
- ▶ **Emission intensity** remains consistent across years within each company, showing little variation.
- ▶ **Top intensity emitters** include **Alpha Metallurgical Resources** and **British Coal Corp**, both above 2.7 MtCO₂e per unit.
- ▶ Major producers like the **Former Soviet Union** and **Saudi Aramco** reached peak outputs in the **1980s–2010s**, sustaining long-term environmental impact.

Top 10 Parent Entities by Average Emission Intensity

Parent Entity	Avg Emission Intensity (MtCO ₂ e per Production Unit)	First Year	Last Year	Total Emissions (MtCO ₂ e)	Total Production (Production Unit)
Alpha Metallurgical Resources	2.829	1981	2022	6127.223	2207.341
American Consolidated Natural Resources	2.714	1988	2022	2239.722	825.205
British Coal Corporation	2.714	1947	1994	19745.361	7275.000
Wolverine Fuels	2.714	2000	2022	384.717	141.746
UK Coal	2.714	1995	2015	881.959	324.950
North Korea	2.713	1962	2022	4103.679	1456.916
Ukraine	2.669	1992	2022	4969.163	1755.868
Whitehaven Coal	2.668	2006	2022	428.392	170.083
Exxaro Resources Ltd	2.668	1998	2022	2159.901	886.440
Vale	2.668	2007	2022	317.152	117.036

Conclusion

- ▶ The data shows that over time, Total Emissions have substantially increased throughout the years. A significant rise began around the 1950s, with emissions skyrocketing in the early 1990s and continuing to rise to the present day. Several major entities have contributed high levels of emissions over the years, one of which was the former Soviet Union, which in the late 1980s produced 5,000 MtCO₂e, making it the highest producer during that period. According to the most recent data from 2022, China (specifically from coal production) is currently the largest producer of emissions, with 12,000 MtCO₂e.
- ▶ How could your team and business apply these insights?
Based on the dataset, the highest recorded Total Emissions occurred in 2022, reaching 35,000 MtCO₂e, which emphasizes the growing environmental challenge.
- ▶ What next steps should you or your stakeholders take based on these findings?
The reduction of Total Emissions year after year is urgently needed. If the current trend continues, there will be severe consequences. Many argue that climate change is one of the direct results of continued air pollution being released each year. This could have a drastic impact on our overall health, both physically and mentally. We must become better stewards of the planet we live on.
- ▶ Is there additional data that could expand on these findings?
Geographical data that identifies the specific sources of emissions worldwide could further enhance the analysis. Additionally, data on the consumption of key commodities for each country would be beneficial for a deeper understanding of the emission drivers.